

FORMALISING THE CONTRACTION MAPPING PRINCIPLE

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INTRODUCTION

The fact that a contraction over a complete metric space has a unique fixed point was first proved by Stefan Banach around 1920, in the context of the normed linear space that bring his name, and, in its most general instance, in 1930 by Renato Caccioppoli. [JJT24]

The principal idea of the proof is that, given a fixed point $x^* = f(x^*)$ for the prescribed contraction $f : X \rightarrow X$, then it is a fixed point of the composition $f \circ f$ too, in fact,

$$f(f(x^*)) = f(x^*) = x^*$$

and thus, iteratively, of the composition

$$f \underbrace{\circ \dots \circ}_{k \text{ times}} f.$$

This means that we can define a sequence $\{x_k\}_{k \in \mathbb{N}}$ given by the iterative update $x_{k+1} = f(x_k)$ so that $x_k \rightarrow x^*$ as $k \rightarrow +\infty$. Completeness of the metric space X is crucial, since x_k is a Cauchy sequence and thus this iterative scheme will always converge to the fixed point of f , uniqueness following from the contractivity assumption. Here, we provide a Lean formalization of the precise proof of this fixed point principle for contraction mappings, mainly following Eberhard Zeidler's "bible" of fixed-point theory. [Zei]

PRELIMINARY DEFINITIONS

First of all, we aim to define the class of Complete Metric Spaces, that is, we let X be a Type, that is equivalent to just being a set, endowed with a nonnegative, nondegenerate, symmetric distance function $d : X \times X \rightarrow \mathbb{R}$, satisfying the triangle inequality: given any $\{x_0, x_1, \dots, x_n\} \subset X$, $d(x_n, x_0) \leq \sum_{i=0}^{n-1} d(x_{i+1}, x_i)$.

```
class CompleteMetricSpace (X : Type) where
  d : X → X → ℝ                                -- distance function
  d_nonneg : ∀ x y, 0 ≤ d x y                     -- non-negativity
  d_eq_zero_iff : ∀ x y, d x y = 0 ↔ x = y       -- identity of indiscernibles
  d_symm : ∀ x y, d x y = d y x                   -- symmetry
  d_triangle : ∀ (n : ℕ) (x : ℕ → X),             -- triangle inequality
    d (x n) (x 0) ≤ Finset.sum (Finset.range n) (fun i => d (x (i + 1)) (x i))
```

By defining the notion of Cauchy sequence,

```
def cauchy_seq (x : ℕ → X) : Prop :=
  ∀ ε : ℝ, ε > 0 → ∃ N : ℕ, ∀ m n, m ≥ N → n ≥ N → d (x m) (x n) < ε
```

and of convergence in the topology induced by the prescribed metric d ,

```
def convergence (x : ℕ → X) (lim : X) : Prop :=
  ∀ ε : ℝ, ε > 0 → ∃ N : ℕ, ∀ n, n ≥ N → d (x n) (lim) < ε
```

we need just to add the completeness axiom to our class description.

```
completeness : ∀ x : ℕ → X, cauchy_seq d x → ∃ lim, convergence d x lim
```

Moreover, we take the following sequential characterization

$$\lim_{n \rightarrow +\infty} f(x_n) = f\left(\lim_{n \rightarrow +\infty} x_n\right)$$

as our definition of continuity,

```
def continuity (f : X → X) : Prop := ∀ (x : ℕ → X) (lim : X),
  convergence CompleteMetricSpace.d x lim →
  convergence CompleteMetricSpace.d (fun n right f (x n)) (f lim)
```

and we shall say that such a continuous map is a contraction if it is Lipschitz:

$$d(f(x), f(y)) \leq L d(x, y), \quad x, y \in X$$

for some constant $L \in [0, 1)$.

THE PROOF

In the following, (X, d) is a complete metric space.

We fix a contraction $f : X \rightarrow X$ and some initial point $x_0 \in X$ and define inductively the so-called Peano fixed-point iteration:

```
def Peano_iterates (f : X → X) (x0 : X) : ℕ → X
- 0      right x0                                -- Base case: x (0) = x0
- (n + 1) right f (Peano_iterates f x0 n)        -- Inductive step: x (n+1) = f (x (n))
```

INDUCTION LEMMA. For all $n \geq 0$,

$$d(x_{n+1}, x_n) \leq L^n d(f(x_0), x_0).$$

Proof. We argue by induction on $n \geq 0$.

The base case $n = 0$:

$$d(x_1, x_0) \leq L^0 d(f(x_0), x_0)$$

is trivial, as it is enough to rewrite repeatedly the definition of Peano iterates for $n = 0$ to get

$$d(f(x_0), x_0) \leq d(f(x_0), x_0).$$

For the inductive step, we have the following goal:

$$d(x_{n+2}, x_{n+1}) \leq L^{n+1} d(f(x_0), x_0)$$

given the inductive assumption

```
ih : CompleteMetricSpace.d (Peano.iterates f x0 (n + 1)) (Peano.iterates f x0 n) ≤
    L ^ n * CompleteMetricSpace.d (f x0) (x0)
```

and in particular we can conveniently rewrite our goal as:

$$d(f(f(x_n)), f(x_n)) \leq L \cdot L^n d(f(x_0), x_0)$$

which is true because we can insert in the middle:

$$d(f(f(x_n)), f(x_n)) \leq L d(f(x_n), x_n) \leq L \cdot L^n d(f(x_0), x_0)$$

which we do in the following two steps

have step1 :

```
CompleteMetricSpace.d (f (f (Peano.iterates f x0 n))) (f (Peano.iterates f x0 n))
  ≤ L * CompleteMetricSpace.d (f (Peano.iterates f x0 n)) (Peano.iterates f x0 n) := by
  exact hcontraction (f (Peano.iterates f x0 n)) (Peano.iterates f x0 n)
```

i.e. we have the first inequality by the contractivity assumption, and

have step2 :

```
L * CompleteMetricSpace.d (f (Peano.iterates f x0 n)) (Peano.iterates f x0 n)
  ≤ L * (L^n * CompleteMetricSpace.d (f x0) (x0))
  apply mul_le.mul_of_nonneg_left
  exact ih
  exact hL.1
```

that is, for the second inequality we cancel the L 's on both sides, so that by the inductive hypothesis we reduce to $0 \leq L$, which we assumed. $\square$

GEOMETRIC BOUND LEMMA. For all $n \geq 0$ and $m \geq 1$,

$$d(x_{n+m}, x_n) \leq \frac{L^n}{1+L} d(f(x_0), x_0).$$

Proof. We define the sequence $\mathcal{P}_i = x_{n+i}$ for $i \geq 0$, so that

$$\begin{aligned} d(\mathcal{P}_m, \mathcal{P}_0) &\leq \sum_{i=0}^{m-1} d(\mathcal{P}_{i+1}, \mathcal{P}_i) && \text{(triangle inequality)} \\ &\leq \sum_{i=0}^{m-1} L^{n+i} d(f(x_0), x_0) && \text{(induction lemma)} \\ &\leq \frac{L^n}{1+L} d(f(x_0), x_0) && \text{(geometric series)} \end{aligned}$$

which is, in Lean, exactly the following:

calc

```
CompleteMetricSpace.d (P m) (P 0)
  ≤ Finset.sum (Finset.range m) (fun i => CompleteMetricSpace.d (P (i + 1)) (P i)) := by
  exact CompleteMetricSpace.d.triangle m P
- ≤ Finset.sum (Finset.range m) (fun i => L^(n + i)) * CompleteMetricSpace.d (f x0) (x0) := by
  rw [Finset.sum_mul] -- distribute sum over multiplication
  apply Finset.sum_le_sum -- use that d(f^{n+i+1} x0, f^{n+i} x0) ≤ L^{n+i} * d(f x0, x0)
  intro i - -- for each term in the sum, apply the previous lemma:
  apply induction f x0 hL hcontraction (n + i)
- ≤ (L^n) * (1 / (1 - L)) * CompleteMetricSpace.d (f x0) (x0)
```

while applying the usual bound of the geometric series is a bit more delicate, as we are actually performing the step:

$$d(f(x_0), x_0) \sum_{i=0}^{m-1} L^{n+i} \leq d(f(x_0), x_0) \frac{L^n}{1+L}$$

involving an infinite sum. In fact, we can apply the following:

```
mul_le_mul_of_nonneg_right {a b c} (hbc : b ≤ c) (ha : 0 ≤ a) : b * a ≤ c * a
```

so that we reduce to proving that $0 \leq a \equiv d(f(x_0), x_0)$ which follows from the nonnegativity axiom of the prescribed metric, together with checking

$$b \equiv \sum_{i=0}^{m-1} L^{n+i} \leq \frac{L^n}{1+L} \equiv c$$

so that our claim can follow.

Of course,

```

apply mul.le.mul.of_nonneg.right      -- cancel d(f x0, x0) on both sides
· simp_rw [pow.add]                  -- rewrite L^(n+i) = L^n * L^i
rw [← Finset.mul.sum]                -- factor ∑ i L^(n+i) = L^n * ∑ i L^i
apply mul.le.mul.of_nonneg.left      -- cancel L^n on both sides

```

so we reduce to proving

$$\sum_{i=0}^{m-1} L^i \leq \frac{1}{1-L}.$$

Here, we need to use the formula for the geometric series, as follows:

$$\sum_{i=0}^{m-1} L^i = \frac{L^m - 1}{L - 1} = \frac{1 - L^m}{1 - L} \leq \frac{1}{1 - L}$$

precisely,

```

rw [geom.sum.eq (ne.of.lt hL.2)]      -- use formula for geometric series
rw [← neg.div.neg.eq, neg.sub, neg.sub] -- rearrange signs
rw [div.eq.mul.inv, div.eq.mul.inv]    -- rewrite div. as multiplication by inv.
apply mul.le.mul.of_nonneg.right      -- cancel (1 - L)^-1 on both sides
· linarith [pow.nonneg hL.1 m]

```

and we are done because $L \geq 0$, so $L^m \geq 0$ and $1 - L^m \leq 1$. □

LEMMA. The Peano iterates form a Cauchy sequence.

Proof. We fix $\epsilon > 0$, so we have to prove

$$\exists N \text{ s.t. } \forall m, n \geq N : d(x_m, x_n) < \epsilon.$$

Because $0 < L < 1$, $L^n \rightarrow 0$ as $n \rightarrow +\infty$ and thus, letting

$$C \equiv \frac{1}{1-L} d(f(x_0), x_0)$$

also $L^n C \rightarrow 0$. We prove this as a side fact:

```

have h.tendsto : Filter.Tendsto (fun n => L^n * C) Filter.atTop (nhds 0) := by
  simpa using (tendsto.pow_atTop_nhds_zero_of_lt_one hL.1 hL.2).mul.const C

```

and now, for the delicate part, namely that there must be some index N such that for all $n \geq N$, the value $L^n C$ is smaller than our specific ϵ , we need to substitute the “ ϵ, N ” definition of convergence,

```

rw [Metric.tendsto_atTop] at h.tendsto

```

```

obtain ⟨N, hN⟩ := h_tendsto ε hε
exists N

```

whence, in particular, we need to show that, given any two indices $p, q \geq N$: $d(x_p, x_q) < \epsilon$, assuming without loss of generality that $p \leq q$ so that we can let $m \equiv q - p$ so that by the geometric bound above:

$$d(x_p, x_q) = d(x_p, x_{p+m}) \leq \frac{L^p}{1-L} d(f(x_0), x_0) = L^p C$$

```

have bound := geometric_bound f hL hcontraction x0 p m
-- by symmetry, d(x_{p+m}, x_p) ≤ L^p * C
rw [CompleteMetricSpace.d.symm]

```

but we already know that $L^p C < \epsilon$ so the desired claim follows by transitivity:

```

apply lt_of_le_of_lt bound

```

and Lean leaves us with the goal of proving that $L^p C \geq 0$, which is trivial because every term in the expression is nonnegative. □

Now, we are finally ready to prove existence and uniqueness of fixed points.

For existence, we make the following assumptions:

```

theorem fixed_point (x0 : X) (hL : 0 ≤ L ∧ L < 1) (hcontraction : contraction L f)
(limit_unique : ∀ {x : ℕ → X} {a b : X},
convergence CompleteMetricSpace.d x a → convergence CompleteMetricSpace.d x b → a = b)
(limit_shift : ∀ {x : ℕ → X} {l : X},
convergence CompleteMetricSpace.d x l → convergence CompleteMetricSpace.d (fun n => x (n + 1)) l)
(f_continuous : ∀ {x : ℕ → X} {l : X},
convergence CompleteMetricSpace.d x l → convergence CompleteMetricSpace.d (fun n => f (x n)) (f l))
: ∃ x_fixed : X, f x_fixed = x_fixed

```

namely that:

- limits of sequences, in the topology induced by the prescribed metric d , are unique;
- if, as $n \rightarrow +\infty$, $x_n \rightarrow \ell$ then so does the “shifted” sequence $x_{n+1} \rightarrow \ell$;
- the fixed map $f : X \rightarrow X$ is continuous, where again we take the transference principle:

$$\lim_{n \rightarrow +\infty} f(x_n) = f\left(\lim_{n \rightarrow +\infty} x_n\right)$$

as our definition of continuity.

Now, the proof of existence can be condensely written as:

$$f(x^*) = \lim_{n \rightarrow +\infty} f(x_n) = \lim_{n \rightarrow +\infty} x_{n+1} = x^*$$

however, a lot is unfolding within this calculation. This is its formalisation:

```

-- We consider the Cauchy sequence of Peano iterands
have h_cauchy := peano_seq_is_cauchy f x0 hL hcontraction
-- By completeness, we extract a limit x_bar
rcases CompleteMetricSpace.completeness (Peano.iterates f x0) h_cauchy with ⟨x_bar, h_lim⟩
-- W.t.s. x_bar is a fixed point
exists x_bar
-- Since f is continuous and x.n → x_bar, it follows that x_{n+1} = f(x.n) → f(x_bar)
have h_left : convergence CompleteMetricSpace.d (fun n => f (Peano.iterates f x0 n)) (f x_bar) :=
  f_continuous h_lim
-- But x_{n+1} → x_bar
have h_right : convergence CompleteMetricSpace.d (fun n => Peano.iterates f x0 (n + 1)) x_bar :=
  limit_shift h_lim
-- By uniqueness of limits, f(x_bar) = x_bar
apply limit_unique h_left h_right

```

Now, we conclude with the uniqueness part. We only leverage the contraction hypotesis: in fact, given two fixed points $x = f(x)$ and $y = f(y)$, then

$$d(x, y) = d(f(x), f(y)) \leq Ld(x, y) \iff x = y.$$

This concludes a full formalization for the proof of the following theorem.

BANACH-CACCIOPPOLI FIXED POINT THEOREM. A contraction of a complete metric space has a unique fixed point.

DIFFICULTIES AND CONCLUSIONS

When I decided to formalise the Banach fixed point theorem, the first challenge was to have complete metric spaces to work with in Lean. At first, I tried to import the relevant sections of Mathlib.Topology.MetricSpace. However, that started to feel rather cumbersome from the very first moments, and I liked the idea of my Lean file to be self-contained. So I went for having a streak of definitions and variable declarations. The problem with such approach was that I could hardly access them whenever needed, for instance inside a proof environment. Luckily around that time we saw, during a lecture, the ‘class’ method, which ultimately I decided to use due to its convenience. For simplicity, I put as axioms the generalized triangle inequality (for sequences, rather

than deriving it from the classical one involving just a couple of points) and the Cauchy characterization of completeness.

The first proof was an easy argument by induction that one would not even bother to write down on a blackboard. However, the formalisation was kind of instructing to me as, in the induction step, I found myself with a goal not really involving the terms present within the inductive hypothesis. What I needed to do was to insert a middle term in the inequality, and prove the two bounds separately. Not really knowing how to do it, I asked Google Gemini to find a me a tactic that I could find useful and he suggested me to divide the goal into a couple of steps with the ‘have’ tactic. I was similarly suggested the ‘calc’ tactic for the proof of the geometric bound, which also simplified the inequality of my objective greatly and allowed me to formalize the streak of inequalities a human would perform in the informal proof. Another minor technical challenge, which I easily overcame in the formalisation of this step, was the application of the triangle inequality for the following bound

$$d(x_{n+m}, x_n) \leq \sum_{i=0}^{m-1} d(x_{n+i+1}, x_{n+i})$$

which Lean did not recognize until the introduction of the auxiliary sequence $\mathcal{P}_i = x_{n+i}$ so that the goal matched exactly the prescribed formulation of the triangle inequality axiom. The rest of the proof consisted of trivial manipulations of the geometric series, and other finite sums and multiplications, the relevant tactics to perform which I was readily suggested by LeanDex. Also, I initially wanted to use a direct bound involving the infinite geometric series, but that did not work for me and I later succeed in rewriting the calculation and to formalize it by means of the formula for the partial geometric sum.

What remained to actually prove was that this bound implies that the Peano iterates form a Cauchy sequence: again, a professor of mathematical analysis writing the proof on a blackboard would not bother to explain it, as it is evident from the inequality obtained. I tried to prove it using the definition, but that was actually impossible to do without taking limits. Indeed, also informally, one says:

$$d(x_{n+m}, x_n) \leq \frac{L^n}{1-L} d(f(x_0), x_0) \lesssim L^n \longrightarrow 0 \quad \text{as} \quad n \rightarrow +\infty$$

because $L < 1$ and, since I did not import anything specific from MathLib, I decided to leave this proof sorried for the moment as it did not seem immediate to me how to render the limiting behaviour using ϵ ’s. Later, I looked into Mathlib.Analysis and imported the relevant material to take limits: rather nicely, I once more proved an inequality by inserting a middle term, so that I could in part adapt a similar idea to the first proof. Also, I learned the rather intuitive ‘wlog’ tactic.

The last effort, was to find a way to make sense of the existence portion of the proof of the theorem, where one implicitly uses continuity and uniqueness of limits (and also the informally trivial fact that the subsequence x_{n+1} has the same limit of the sequence x_n). I felt that proving these facts was out of the scope of the project, also

being facts that one could easily find in the relevant sections of MathLib if imported. So, I decided to assume them as variables.

In conclusion, when I finished the project and everything in my file compiled, I had the impression of having understood the proof of the theorem at its fullest. I have seen the theorem in a couple of courses I have taken in my life, and I always kind of skimmed over the seemingly straightforward proof. But this formalisation showed how nicely the few assumptions (contractivity, completeness and axioms of metric space) are essential in each step of the proof, something that, while reading the proof from lecture notes or a textbook, I did not see explicitly stated and accordingly did not realize when looking at automatic computations similar to many others in analysis.

REFERENCES

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